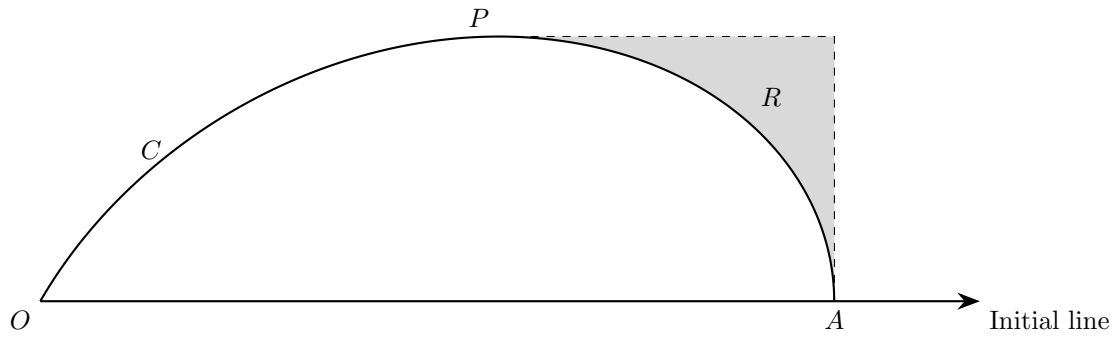


## Questions

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1. The curve  $C$  shown in the figure above has polar equation

$$r = 1 + 2 \cos 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the point  $P$  on  $C$ , the tangent to  $C$  is parallel to the initial line.

- (a) Show that  $P$  has polar coordinates  $\left(2, \frac{\pi}{6}\right)$ . [4]

The point  $A$  on  $C$  has polar coordinates  $(3, 0)$ .

The finite region  $R$ , shown shaded in the figure above, is bounded by the arc  $AP$  of  $C$ , the tangent to  $C$  at  $P$  and the line through  $A$  parallel to  $\theta = \frac{\pi}{2}$ .

- (b) Find the exact area of  $R$ . [9]

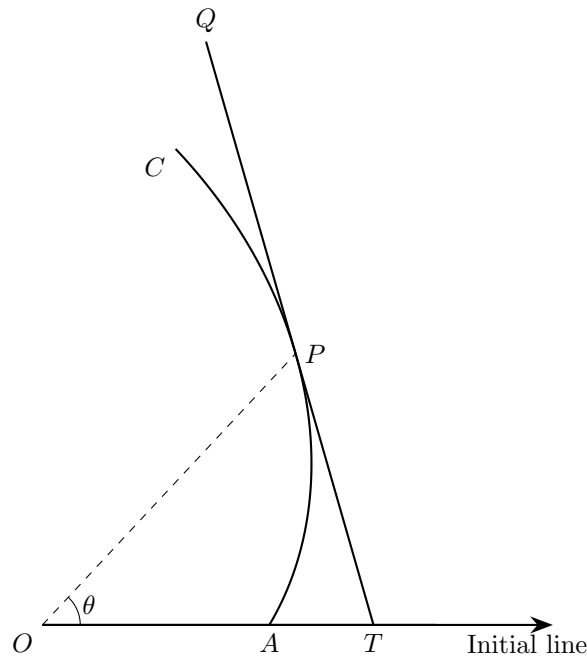
2. The curve  $C$  has polar equation

$$r = 2 \sin 3\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the non-pole point  $P$  on  $C$ , the tangent to  $C$  is parallel to the initial line.

Given that  $O$  is the pole, find the exact length of the line  $OP$ .

[7]



3. The diagram shows part of a spiral  $C$ .

The point  $P$  has polar coordinates  $(r, \theta)$ , where  $0 \leq \theta \leq \frac{\pi}{2}$ .

The tangent at  $P$  is the line  $TPQ$ . The point  $A$  lies on  $C$  and on the initial line, with  $OA = 2$ .

Let  $\psi$  be the angle that  $TPQ$  makes with the initial line.

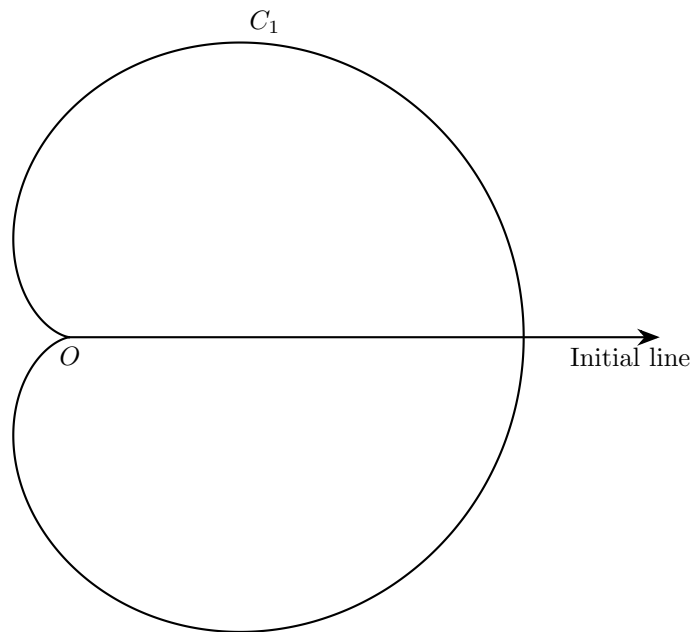
(a) By expressing the Cartesian coordinates of  $P$  in terms of  $r$  and  $\theta$ , show that

$$\tan \psi = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} \quad [4]$$

(b) The spiral is equiangular: at every point  $P$ , the acute angle between  $OP$  and  $TPQ$  is a constant  $\alpha$ , where  $0 < \alpha < \frac{\pi}{2}$ .

Use the result of part (a) to show that the polar equation of  $C$  is

$$r = 2e^{(\cot \alpha)\theta} \quad [7]$$



4. The diagram shows the sketch of a curve  $C_1$ .

The polar equation of the curve  $C_1$  is

$$r = 2(1 + \cos \theta), \quad -\pi \leq \theta \leq \pi$$

- (a) Find the area of the region bounded by the curve  $C_1$ . [5]

- (b) The curve  $C_2$  whose polar equation is

$$r = \sin \theta, \quad 0 \leq \theta \leq \pi$$

intersects the curve  $C_1$  at the pole  $O$  and at the point  $A$ . The straight line drawn through  $A$  perpendicular to the initial line intersects the part of  $C_1$  above the initial line again at the point  $B$ .

- (i) Find the polar coordinates of  $A$ . [4]  
 (ii) Show that the length of  $OB$  is  $\frac{6}{5}$ . [6]  
 (iii) Find the length of  $AB$ , giving your answer to three significant figures. [3]

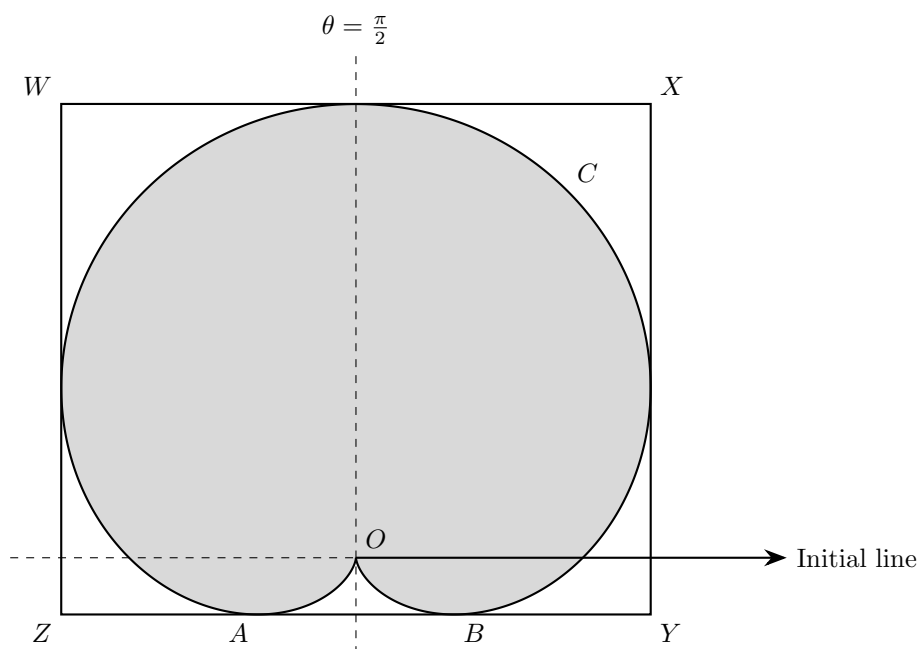
5. The curve  $C$  has polar equation

$$r = 3 \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

At the non-pole point  $P$  on  $C$ , the tangent to  $C$  is perpendicular to the initial line.

Given that  $O$  is the pole, find the exact length of the line  $OP$ .

[7]



6. The figure shows a sketch of the cardioid  $C$  with equation

$$r = a(1 + \sin \theta), \quad 0 \leq \theta < 2\pi$$

Also shown are the tangents to  $C$  that are parallel and perpendicular to the initial line. These tangents form a rectangle  $WXYZ$ .

(a) Find the area of the finite region, shaded in the figure, enclosed by  $C$ . [6]

(b) Find the values of  $\theta$  for which the tangent to  $C$  is parallel to the initial line. Hence find the polar coordinates of the points  $A$  and  $B$ , where  $ZY$  touches  $C$ . [6]

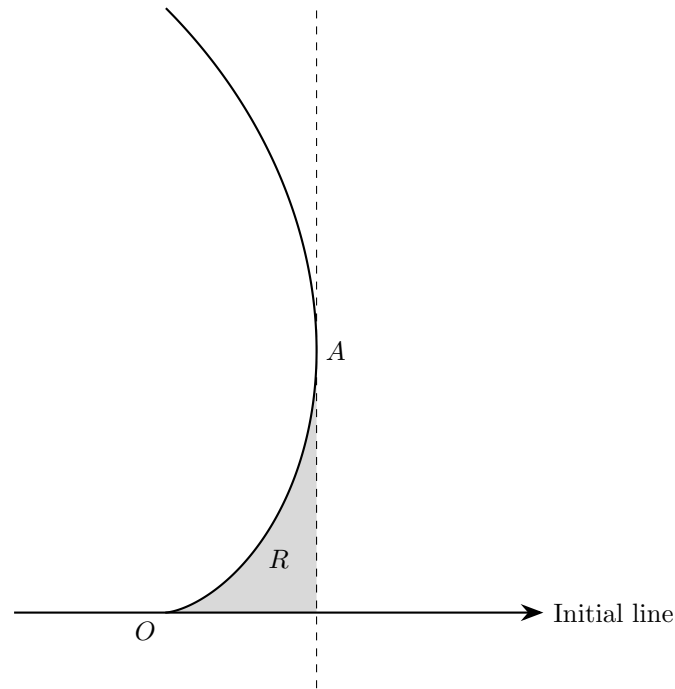
(c) Hence show that the length of  $WZ$  is  $\frac{9a}{4}$ . [2]

Given that the length of  $WX$  is  $\frac{3\sqrt{3}a}{2}$

(d) find the area of the rectangle  $WXYZ$ . [2]

A logo is modelled by the cardioid  $C$ , where  $a = 8$  cm. The logo is cut from the rectangular card  $WXYZ$ , shown in the figure.

(e) Find a numerical value for the area of card wasted in making the logo. [3]



7. The diagram above shows a sketch of the curve  $C$  with equation

$$r = 4(1 - \cos \theta) \quad 0 \leq \theta \leq \frac{\pi}{2}$$

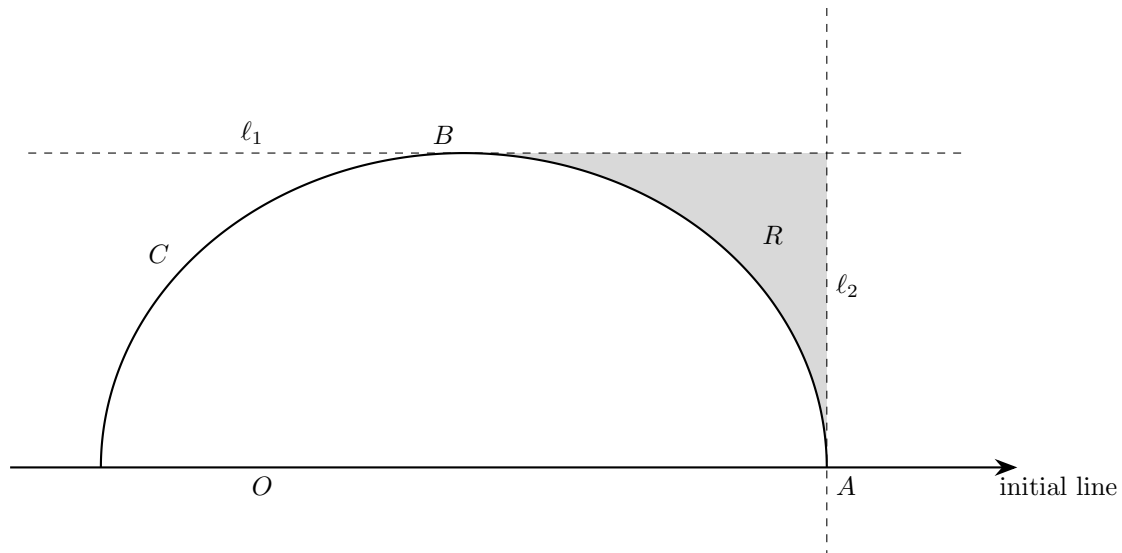
The diagram above also shows the tangent to  $C$  at the point  $A$ , where  $A$  is not the pole. This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of  $A$  are  $\left(2, \frac{\pi}{3}\right)$ . [4]

The finite region  $R$ , shown shaded in the diagram above, is bounded by  $C$ , the tangent at  $A$  and the initial line.

(b) Use calculus to show that the exact area of  $R$  is  $\frac{1}{2}(15\sqrt{3} - 8\pi)$ . [6]





8. The figure above shows a sketch of the curve  $C$  with equation

$$r = \frac{6}{2 - \cos \theta} \quad 0 \leq \theta \leq \pi$$

Given that  $C$  meets the initial line at the point  $A$ , as shown in the figure above

(a) write down the polar coordinates of  $A$ .

[1]

The line  $\ell_1$ , also shown in the figure above, is the tangent to  $C$  at the point  $B$  and is parallel to the initial line.

(b) Use calculus to determine the polar coordinates of  $B$ .

[4]

The line  $\ell_2$ , also shown in the figure above, is the tangent to  $C$  at  $A$  and is perpendicular to the initial line.

The region  $R$ , shown shaded in the figure above, is bounded by  $C$ ,  $\ell_1$  and  $\ell_2$ .

(c) By first obtaining a Cartesian equation for  $C$ , use algebraic integration to find the exact area of  $R$ , giving your answer in the form  $\sqrt{3}(p + q\pi)$  where  $p$  and  $q$  are constants to be determined.

[8]